Project — Online Learning

Sequential forecasting, decision under uncertainty and confidence intervals

Paul Minchella Groups of 3 or 4 students Academic year 2026/2027

Project guidelines

- ▷ Groups of **3 or 4** students.
- ▷ Written **in English or in French**.
- ▷ Topic choice to be sent before **29/11/2026, 11:59 pm**.
- ▷ Final submission before **20/12/2026**, to be sent to my email `paul.minchella@univ-lyon2.fr`.
- ▷ Clear file name, such as `name1_name2_name3.pdf`.
- ▷ Main deliverables: a **written report** and an executed **notebook** (or **illustrative code**, a **GitHub** link, an HTML page, or any other equivalent format).

I remain available by email for any questions you may have, whatever their nature.

1. Objective

Each group chooses a **public dataset** (a time series or a panel of series) and implements a complete **online learning** pipeline: data arrive one at a time, the model predicts $\hat{y}_t$ *before* observing y_t, incurs a loss $\ell_t(\hat{y}_t, y_t)$, and then updates itself. The goal is to **compare at least three online methods**, to **quantify the uncertainty** of this comparison with an **empirical bootstrap** on the final metrics and, as a bonus, to build **online conformal prediction intervals** (ACI).

2. Schedule and deliverables



2.1 Step 1 — Scoping note (no later than 29/11/2026, 11:59 pm)

A **short and concise** document (Word, Overleaf, . . . , 1 to 2 pages) presenting:

- ▷ the chosen **dataset**: source, period, frequency, target variable, possible covariates;
- ▷ a few **descriptive statistics**: trend, seasonalities, missing values, zeros, breaks, distribution of the target;
- ▷ the **models to be tested** and the precise **task** (horizon, granularity, target quantile τ if relevant);
- ▷ the **evaluation** protocol (metrics, test period) and the expected **interpretations**.

2.2 Step 2 — Final report (no later than 20/12/2026)

A **report written on Overleaf**, together with a **cleanly executed notebook** (or a GitHub repository, an HTML page, or any other equivalent deliverable of your choice), reproducible end to end.

2.3 Step 3 — Oral presentation (if possible, during January)

Each group of 3 or 4 students presents its results, followed by questions.

3. Suggested outline of the report

1. **Introduction and motivation**: why this problem, why an *online* setting?
2. **Presentation of the dataset and of the task**: description, preprocessing, sequential protocol.
3. **Learning with at least 3 online methods** among: FTL, FTRL, OGD, regression OGD, penalized regression OGD, Newsvendor (OGD on the pinball loss).
   Bonus: other models (Trend-Season, ARMA, SARIMA, boosting, neural networks, ...).
4. **Metrics and interpretations**: model comparison, regret, analysis of errors over time.
5. **Empirical confidence intervals** (bootstrap) on the chosen metric(s).
6. Bonus: **conformal prediction intervals with ACI**.
7. **Conclusions, benefits and limitations**.

Reminder of the methods

Method	Update
FTL	$x_{t+1} = \arg\min_{x\in\mathcal{K}} \sum_{s=1}^{t} \ell_s(x)$
FTRL	$x_{t+1} = \arg\min_{x\in\mathcal{K}} \left\{ \sum_{s=1}^{t} \ell_s(x) + \frac{1}{\eta} R(x) \right\}$
OGD	$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla \ell_t(x_t))$
Regression OGD	$\hat{y}_t = \theta_t^\top z_t, \quad \theta_{t+1} = \theta_t - \eta_t(\hat{y}_t - y_t)\, z_t$
Penalized OGD	$\theta_{t+1} = \theta_t - \eta_t \left[(\hat{y}_t - y_t)\, z_t + \lambda \theta_t\right]$ (ridge; or ℓ_1 proximal step)
Newsvendor	$q_{t+1} = q_t - \eta_t \left(\mathbf{1}\{y_t < q_t\} - \tau\right), \quad \tau = \frac{c_b}{c_h + c_b}$

> **Remark.** A naive baseline (last value, or seasonal naive $\hat{y}_t = y_{t-s}$) is strongly recommended: it sets the scale from which a method genuinely "learns" something.

4. Evaluation: metrics and empirical bootstrap

4.1 Metrics

Over the test period $t = 1, \ldots, T$, each model m produces a sequence of losses $e_t^{(m)}$. Depending on the task:

▷ **point accuracy**: MAE $= \frac{1}{T}\sum_t |y_t - \hat{y}_t|$, RMSE, MASE;

▷ **quantile / decision**: pinball loss $\rho_\tau(y_t - q_t)$ and newsvendor cost $c_h(q_t - y_t)_+ + c_b(y_t - q_t)_+$, stockout rate $\frac{1}{T}\sum_t \mathbf{1}\{y_t > q_t\}$;

▷ **cumulative regret**: $\mathrm{Reg}_T = \sum_{t=1}^T \ell_t(x_t) - \min_{x\in\mathcal{K}} \sum_{t=1}^T \ell_t(x)$.

4.2 Empirical bootstrap on the final bar chart

Expected

The **final bar chart** comparing the models must display, for each bar, a $1-\alpha$ **bootstrap confidence interval** (e.g. 95 %) on the metric shown. State on which metric(s) you compute it and why.

Since the losses $e_t^{(m)}$ are **serially dependent**, resample by **blocks** (circular block bootstrap) rather than point by point:

1. choose a block length L (order of magnitude: $L \approx T^{1/3}$, or the seasonal period);
2. for $b = 1, \ldots, B$ (e.g. $B = 1000$): draw block starting points uniformly, concatenate blocks of length L (wrapping around the end of the series) until T indices are obtained, then compute the bootstrap metric $M_m^{*(b)}$;
3. percentile interval: $\Big[\, q_{\alpha/2}\big(M_m^*\big)\,,\; q_{1-\alpha/2}\big(M_m^*\big)\,\Big]$.

> **Remark.** Use the **same resampled indices for all models** (paired bootstrap). You can then compute an interval on the *difference* $M_m - M_{\text{ref}}$: if this interval contains 0, the superiority of model m over the baseline is not established.

BONUS — Online Conformal Prediction: ACI

Build online a prediction interval around $\hat{y}_t$, with target level $1-\alpha$. With the score $s_t = |y_t - \hat{y}_t|$ and $\hat{Q}_t(p)$ the empirical quantile of order p of past scores (possibly over a rolling window):

$$C_t = \Big[\, \hat{y}_t - \hat{Q}_t(1-\alpha_t)\,,\; \hat{y}_t + \hat{Q}_t(1-\alpha_t)\,\Big], \qquad \alpha_{t+1} = \alpha_t + \gamma\Big(\alpha - \mathrm{err}_t\Big), \quad \mathrm{err}_t = \mathbf{1}\{y_t \notin C_t\}.$$

After a miss, α_t decreases and the interval widens; after a hit, it shrinks.

▷ To report: **empirical coverage** $\frac{1}{T}\sum_t(1-\mathrm{err}_t)$, **rolling coverage**, **mean width** of the intervals, and sensitivity to the step size γ.

▷ Reference: Gibbs & Candès (2021), *Adaptive Conformal Inference Under Distribution Shift*, NeurIPS.

5. Suggested datasets

Indicative list: you may choose another public dataset, provided you justify it in the scoping note. The first three are **demand / inventory** datasets, naturally suited to the newsvendor problem.

#	Dataset	Content	Source
Demand and inventory			
1	M5 Forecasting (Walmart)	Daily sales per product and store, prices, calendar, product hierarchy	https://www.kaggle.com/competitions/m5-forecasting-accuracy
2	Corporación Favorita	Daily sales of a grocery chain, promotions, perishable products	https://www.kaggle.com/competitions/favorita-grocery-sales-forecasting
3	FreshRetailNet-50K	50,000 hourly series of fresh products, annotated stockouts (censored demand)	https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K
Sales and retail			
4	Rossmann Store Sales	Daily sales of more than 1,000 stores, promotions, holidays	https://www.kaggle.com/competitions/rossmann-store-sales
5	Online Retail II (UCI)	Transactions of a UK online retailer, 2009–2011, to be aggregated by day/week	https://archive.ics.uci.edu/dataset/502/online+retail+ii
Energy, mobility, environment			
6	Electricity Load Diagrams (UCI)	Electricity consumption of 370 clients every 15 min, 2011–2014	https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014
7	Metro Interstate Traffic Volume (UCI)	Hourly traffic on the I-94 interstate (Minneapolis–St Paul), 2012–2018, with weather and holidays	https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume
8	Air Quality (UCI)	Hourly measurements from pollutant sensors (CO, NO_2, ...)	https://archive.ics.uci.edu/dataset/360/air+quality
9	ETT (Electricity Transformer Temperature)	Transformer oil temperature and loads, hourly and 15-min	https://github.com/zhouhaoyi/ETDataset
General benchmarks			
10	Monash Forecasting Repository (M4, Tourism, Traffic, ...)	Academic collection of time-series forecasting benchmarks	https://forecastingdata.org

Remark. The Kaggle datasets (1, 2, 4) require a free account; the UCI datasets are also available in Python via `ucimlrepo`, and FreshRetailNet-50K via `datasets.load_dataset`. For large datasets, work on a subset (a few products or stores, or an aggregated level) and justify this choice. **Caution**: in inventory problems, observed sales are not demand (sales are censored by stockouts).

6. Tips for success

- ▷ **Respect the online setting**: no future information may be used to predict y_t (normalization, hyperparameter selection, ...).
- ▷ **Interpret rather than pile up**: explain *why* and *when* a model wins or loses; discuss the choices of τ, η, L and the limitations.
- ▷ **Polish the deliverable**: readable, captioned figures, a notebook executed end to end, fixed random seeds.